Regression Metrics

- Error
  - MAE
  - MSE
- Accuracy
  - R^2 Score

reg. Score
(sklearn)

Classification Metrics

- Accuracy
  - $y_{actual} == y_{pred}$

True
false
$\vdots$

$$Acc = \frac{\sum True}{\sum True + \sum false}$$

clf. Score
(sklearn)

- other Metrics

Acc
Error

Confusion Matrix

		prediction 0	prediction 1
Actual	0	TN	FP
	1	FN	TP

Recall

$$\text{precision} = \frac{Tp}{Tp + FP} = \frac{Tp\ \text{pred}}{\text{All pos. pred}}$$

$$\text{Accuracy} = \frac{Tp + TN}{Tp + TN + Fp + FN}$$

- TN → True Negative
- Fp → False positive
- FN → False Negative
- Tp → True positive

$$\text{Recall} = \frac{Tp\ \text{pred}}{\text{All pos. Actual}}$$

Harmonic Mean of precision & Recall ← $\text{F1\_score} = 2 \times \frac{\text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}}$

Actual

Class 0 → 85

class 1 → 15

		pred 0	pred 1
Actual	0	80	5
	1	10	5

$precision\_0 = \frac{80}{90}$

$precision = \frac{5}{5+5} = \boxed{50\%}$

↓ fp → precision ↑

$Acc = \frac{80+5}{80+5+10+5} = \boxed{85\%}$

$Recall\_0 = \frac{80}{85}$

$Recall = \frac{5}{5+10} = \boxed{33\%}$

↓ fN → Recall ↑

$f1.score = 2\,\frac{0.5 \times 0.33}{0.5+0.33}$

$= 40\%$

Most frequent ← Dummy clf

(class 0)

		pred 0	pred 1
Actual	0	80	0
	1	20	0

$\boxed{precision = 0\%}$

$\boxed{Acc = 80\%}$

$\boxed{f1\ score = 0}$

$\boxed{Recall = 0\%}$

	precision	recall	f1-score	support
0	0.93	1.00	0.96	13
1	0.00	0.00	0.00	1
accuracy			0.93	14
macro avg	0.46	0.50	0.48	14
weighted avg	0.86	0.93	0.89	14

True Negatives: 13
False Positives: 0
False Negatives: 1
True Positives: 0

7 $Acc = \frac{TP+TN}{TP+TN+FP+FN} = \frac{13}{14} = 0.93$

class 1

1 $precision = \frac{TP}{TP+FP} = \frac{0}{0} = 0$

2 $Recall = \frac{TP}{TP+FN} = \frac{0}{1} = 0$

3 $f1 = 2 \times \frac{Precision \times Recall}{precision + Recall} = 0$

class 0

4 $precision\_0 = \frac{TN}{TN+FN} = \frac{13}{13+1} = 0.93$

5 $Recall\_0 = \frac{TN}{TN+FP} = \frac{13}{13+0} = 1$

6 $f1\_0 = 2 \times \frac{0.93 \times 1}{0.93+1} = 0.96$

	precision	recall	f1-score	support
0	0.93	1.00	0.96	13
1	0.00	0.00	0.00	1
accuracy			0.93	14
macro avg	0.46	0.50	0.48	14
weighted avg	0.86	0.93	0.89	14

support → # Target

Macro Avg

(8) $Avg\_Precision = \frac{0.93 + 0}{2} = 0.46$

(9) $Avg\_Recall = \frac{1+0}{2} = 0.5$

(10) $Avg\_f1 = \frac{0.96 + 0}{2} = 0.48$

Weighted Avg

(11) $W_{avg}\_precision = \frac{0.93 \times 13 + 0 \times 1}{13+1} = 0.86$

(12) $W_{avg}\_Recall = \frac{1 \times 13 + 0 \times 1}{14} = 0.93$

(13) $W_{avg}\_f1 = \frac{0.96 \times 13 + 0 \times 1}{14} = 0.89$

Confusion Matrix

Binary clf → # classes = 2 → 2 x 2

Multi class → # classes = n → n x n

Iris

$$
\begin{array}{cc|ccc}
 & & & \text{pred.} & \\
 & & 0 & 1 & 2 \\
\hline
 & 0 & TN & FP & TN \\
\text{Actual} & 1 & FN & TP & FN \\
 & 2 & TN & FP & TN
\end{array}
$$

cm → 3 x 3

→ ACC

precision

Recall

f1_score

	precision	Recall	f1 score	Support
class 0	P_0			n_0
" 1	P_1			$n1$
" 2	P_2			$n2$

$$\text{Macro\_Avg} = \frac{P_0 + P1 + P_2}{3}$$

$$\text{Weighted\_Avg} = \frac{P_0 \times n_0 + P_1 \times n1 + P_2 \times n_2}{n_0 + n1 + n2}$$

$$\text{Micro\_Avg} = \frac{\Sigma TP}{\Sigma TP + \Sigma FP}$$

precision

precision vs. Recall Tradeoff

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Covid

TP → Actual : +ve , Pred : +ve ✓

TN → Actual : -ve, pred : -ve ✓

FP → Actual : -ve, pred : +ve → OK (Mole Safe)

FN → Actual : +ve, pred : -ve → ⚠

FP ↑ → precision ↓

FN ↓ → Recall ↑

precision vs. Recall Tradeoff

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Safe videos

TP → Actual : +ve , Pred : +ve ✓

TN → Actual : -ve , pred : -ve ✓

FP → Actual : -ve , pred : +ve → ⚠

FN → Actual : +ve , pred : -ve → OK (More Safe)

FP ↓ → precision ↑

FN ↑ → Recall ↓

Confusion Matrix

		pred 0	pred 1
Actual	0	TN	FP
	1	FN	TP

True Positive Rate: $TPR = \dfrac{TP}{TP + FN}$ = Recall (senstivity)

False positive Rate: $FPR = \dfrac{FP}{FP + TN} = 1 - TNR$ Specificity

$= 1 - \dfrac{TN}{FP + TN}$